



Marwadi
University
Marwadi Chandarana Group

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MARWADI UNIVERSITY

FACULTY OF COMPUTER APPLICATIONS

Programming with Python & Problem Solving (05MC0107)

Programming with Python & Problem Solving (05MC0107) Lab Manual (2026-27)

Program: M.C.A.

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INDEX

Lab	Program	Date	Remarks	Signature
1	Write a program to demonstrate basic input and output operations in Python.			
2	Write a program to illustrate the use of different data types and type casting.			
3	Write a program to perform arithmetic relational and logical operations using Python operators.			
4	Write a program to demonstrate string operations including slicing formatting and built-in string functions.			
5	Write a program to create and manipulate lists using indexing slicing and list comprehensions.			
6	Write a program to illustrate the use of tuples and sets with basic operations.			
7	Write a program to create a dictionary and demonstrate dictionary methods and iteration.			
8	Write a program to explain mutable and immutable objects in Python.			
9	Write a program to define and use user-defined functions with different types of arguments.			
10	Write a program to demonstrate recursion using factorial or Fibonacci series.			
11	Write a program to demonstrate conditional statements using if-else and if-elif-else.			
12	Write a program to check whether a number is positive negative or zero using nested conditions.			
13	Write a program to generate a multiplication table using a for loop.			
14	Write a program to find the sum of digits of a number using a while loop.			
15	Write a program to demonstrate the use of break continue and pass statements.			
16	Write a program to iterate over lists strings and dictionaries using loops.			
17	Write a program to demonstrate list dictionary and set comprehensions.			
18	Write a program to illustrate variable scope using local global and nonlocal variables.			
19	Write a program to demonstrate iterators and			



	iterables in Python.			
20	Write a program to generate a sequence of numbers using generator functions and yield keyword.			
21				
22				
23				
24				
25				
26				
27				
28				
29				
30				
31				
32				
33				
34				
35				
36				
37				
38				
39				
40				
41				
42				
43				
44				
45				
46				
47				
48				
49				
50				

Practical No. 1

Write a program to demonstrate basic input and output operations in Python.

Aim/ Objective:

To write a Python program that demonstrates **basic input and output operations**, by accepting a number from the user using `input()` and displaying it using `print()`.

Source Code:

```
# Write a program to demonstrate basic input and output operations in Python.
```

```
a=input("Enter A number:")
```

```
print("The number is :",a)
```

Output:

```
===== RESTART: D:\92600584022\Python\Unit_1\1.py =====  
Enter A number:8  
The number is : 8  
|
```

Conclusion:

Thus, the Python program successfully demonstrates **basic input and output operations**. The program takes a number as input from the user and displays the entered number as output.

Practical No. 2

Write a program to illustrate the use of different data types and type casting.

Aim/ Objective:

To understand different data types in Python and demonstrate type casting by converting values from one data type to another.

Source Code:

```
# Write a program to illustrate the use of different data types and type casting.
```

```
i=10

flt=10.5 text="Hiii" bool1=True

print("int :",i,type(i)) print("float :",flt,type(flt)) print("text :",text,type(text)) print("bool :",bool1,type(bool1))
print() print("Type casting")

a=20 b=float(a) c=str(b) d=bool(c)

print("Integer to float :",b,type(b)) print("Float to String :",c,type(c)) print("String to Bool :",d,type(d))
```

Output:

```
===== RESTART: D:\92600584022\Python\Unit_1\2.py =====
int : 10 <class 'int'>
float : 10.5 <class 'float'>
text : Hiii <class 'str'>
bool : True <class 'bool'>

Type casting
Integer to float : 20.0 <class 'float'>
Float to String : 20.0 <class 'str'>
String to Bool : True <class 'bool'>
```

Conclusion :

The program successfully demonstrates Python data types and type casting.

Practical No. 3

Write a program to perform arithmetic relational and logical operations using Python operators.

Aim/ Objective:

To perform and understand arithmetic, relational, and logical operations using Python operators.

Source Code:

Write a program to perform arithmetic relational and logical operations using Python operators.

```
print("Arithmetic operators") a=10 b=3 print() print("Addition :",a+b) print("Subtraction :",a-b)
print("Multiplication :",a*b) print("Division :",a/b) print("Modulus :",a%b) print("Exponentiation :",a**b)
print("Floor division :",a//b) print()

print("Relational operators") print()

print("Equal to :",a==b) print("Not equal to:",a!=b) print("Less than:",a<b) print("Greater than:",a>b)
print("Less than equal to:",a<=b) print("Greater than equal to:",a>=b)

print() print("Logical operators") print()

print("And :",a>b and b>0) print("Or :",a>b or b>0) print("Not :",not(b>0))
```

Output:



```
===== RESTART: D:\92600584022\Python\Unit_1\3.py =====
```

```
Arithmetic operators
```

```
Addition : 13  
Subtraction : 7  
Multiplication : 30  
Division : 3.3333333333333335  
Modulus : 1  
Exponentiation : 1000  
Floor division : 3
```

```
Relational operators
```

```
Equal to : False  
Not equal to: True  
Less than: False  
Greater than: True  
Less than equal to: False  
Greater than equal to: True
```

```
Logical operators
```

```
And : True  
Or : True  
Not : False
```

Conclusion :

The program successfully illustrates the use of various Python data types and type conversion.

Practical No. 4

Write a program to demonstrate string operations including slicing formatting and built-in string functions.

Aim/ Objective:

To write a Python program that demonstrates **basic input and output operations**, by accepting a number from the user using `input()` and displaying it using `print()`.

Source Code:

Write a program to demonstrate basic input and output operations in Python.

```
a=input("Enter A number:") print("The number is :",a)
```

Output:

```
===== RESTART: D:\92600584022\Python\Unit_1\1.py =====  
Enter A number:8  
The number is : 8  
|
```

Conclusion:

Thus, the Python program successfully demonstrates **basic input and output operations**. The program takes a number as input from the user and displays the entered number as output.